NIVARAN — PLATFORM TECH STACK

Smart Disaster Early Warning, Live GIS & Relocation Decision-Support System

Complete End-to-End Technology Stack, Architecture Pipeline, Spatial Engines & Libraries

VERSION 2.0.0 PROD | DATE: NOVEMBER 2026 | NDMA & SDMA COMPLIANT

1. HIGH-LEVEL SYSTEM TOPOLOGY & ARCHITECTURE

Multi-Tier Decoupled

Architecture Paradigm:

NIVARAN operates as an asynchronous, zero-downtime, spatial-first architecture designed for mission-critical disaster management. The client utilizes an offline-first Progressive Web Application with local IndexedDB queuing (72h expiry). The backend exposes an asynchronous FastAPI REST gateway with JWT Role-Based Access Control (RBAC), driving a dual-persistence layer combining PostgreSQL 14+ / PostGIS 3.0+ for geodetic spatial queries and MongoDB 7.0+ for high-volume document registries and telemetry.

2. FRONTEND CLIENT & PRESENTATION TIER (SPA / PWA)

Web & Field PWA

CORE FRAMEWORK & RUNTIME

- **React** v18.3.1 — Component architecture, Strict Mode, Suspense, Concurrent Hooks
- **TypeScript** v5.6.2 — Strict mode, comprehensive typing for spatial, hazard & demographic models
- **Vite** v6.0.5 — ESM-native build system, fast Hot Module Replacement, optimized chunking
- **React Router DOM** v6.28.0 — Nested routing, declarative route guards, dynamic code splitting

USER INTERFACE, MOTION & THEMING

- **Tailwind CSS** v3.4.17 — Custom disaster tokens, light/dark theme, responsive mobile-to-EOC screens
- **Framer Motion** v11.15.0 — GPU-accelerated page transitions, smooth sliding drawers, modal tabs
- **Lucide React** v0.469.0 — Crisp vector icon system across dashboards and spatial controls
- **PostCSS / Autoprefixer** v8.4 / v10.4 — Cross-browser compatibility and automated prefixing

STATE MANAGEMENT, OFFLINE SYNC & FORMS

- **Zustand** v5.0.2 — Atomic global state store, selective component re-rendering, multi-tab sync
- **TanStack React Query** v5.62.7 — Async server state caching, background refetch, stale-time management
- **idb (IndexedDB)** v8.0.4 — Offline-first database, 72-hour field mutation queue & auto-reconciliation
- **Zod & React Hook Form** v3.24 / v7.54 — Strict runtime schema validation for field incident input forms
- **DOMPurify** v3.4.16 — Context-aware XSS sanitization for citizen alerts and dynamic map tooltips

3. GEOSPATIAL & LIVE GIS MAPPING STACK

2D / 3D Spatial Engines

MAP RENDERING ENGINES & MULTI-VIEWPORTS

- **MapLibre GL JS** v6.13.0 — High-performance WebGL 2D/3D GPU vector & raster map rendering engine
- **Leaflet & React-Leaflet** v1.9.4 / v4.2.1 — Situational secondary viewports, mini-maps & fast raster overlays
- **AWS Terrarium DEM** 30m Resolution — 3D terrain elevation meshes, slope gradient analysis & flood simulator

SPATIAL COMPUTATIONS, GEOFENCING & VECTORS

- **Turf.js Suite** `v7.4.0` — @turf/buffer, area, distance, centroid, along, nearest-point, polygon checks
- **Supercluster** `v9.1.0` — High-density spatial clustering for 10,000+ villages, shelters & resources
- **Mapbox GL Draw & Geoman** `v1.5.2 / v2.17` — Interactive polygon digitization, boundary editing & hazard containment
- **shpjs** `v6.2.0` — Client-side ESRI Shapefile (.shp/.dbf/.prj) parsing directly into GeoJSON

4. BACKEND API SERVICES & DATA PIPELINE

Python Async Engine

APPLICATION SERVER & FRAMEWORK

- **Python** `v3.10+ / 3.14` — High-performance asynchronous runtime for mathematical risk modeling
- **FastAPI** `v0.115.0+` — Async RESTful microservice framework with OpenAPI/Swagger interactive docs
- **Uvicorn** `v0.30.0+` — Lightning-fast ASGI web server with worker management and keep-alive
- **Pydantic** `v2.8.0+` — Strict data serialization, request/response models & contract validation
- **HTTPX** `v0.27.0+` — Non-blocking async HTTP client for external weather and seismic streams

SECURITY, RATE LIMITING & AUTHENTICATION

- **JWT & Python-JOSE** `HS256` — Stateless access tokens, 120-minute expiry, refresh token rotation
- **Passlib & Bcrypt** `v4.2.0` — Salted cryptographic password hashing with 12 computational rounds
- **Custom Rate Limiter** `Token Bucket` — Per-IP leaky-bucket rate limiting preventing DoS attacks on telemetry APIs
- **RBAC Access Control** `10 Roles` — Granular permissions: Admin, SDMA, District, Field, Responder, Citizen

5. DATABASE & PERSISTENCE INFRASTRUCTURE

Dual-Persistence

POSTGRESQL + POSTGIS (SPATIAL & ADMINISTRATIVE CORE)

- **PostgreSQL** `v14+` — ACID relational database engine for administrative records and users
- **PostGIS Extension** `v3.0+` — Geospatial indexing (GIST R-Tree), EPSG:4326 MultiPolygon geometry
- **Spatial SQL Functions** `ST_Suite` — ST_Intersects, ST_Buffer, ST_Area, ST_Within, ST_DWithin geoprocessing
- **5-Level Admin Hierarchy** `Standard` — Country (India) -> States -> Districts -> Talukas -> Villages

MONGODB & CLIENT-SIDE CACHING

- **MongoDB** `v7.0+` — 18 collections: risk zones, telemetry logs, IDRN resources, scenario runs
- **PyMongo & 2dsphere** `v4.9.0` — Geospatial 2dsphere proximity indexes and GeoJSON polygon serialization
- **MemoryCollection Fallback** `Built-in` — Resilient in-memory fallback guaranteeing 100% API uptime if DB fluctuates
- **IndexedDB (Browser)** `HTML5` — Local client storage for offline mutations, draft reports & basemaps

6. ANALYTICS, REPORTING & PROTOCOL STANDARDS

Open Interoperability

VISUALIZATION, DOCUMENTS & EXPORTING

- **Recharts** `v2.15.0` — Interactive charts: hazard histograms, radar vulnerability & bar metrics
- **jsPDF & html2canvas** `v2.5.2 / v1.4.1` — Client-side generation of signed Government Orders & SitRep PDF files
- **PapaParse** `v5.4.1` — Streaming parsing and export of IDRN inventory and census records

EXTERNAL INTEGRATIONS & PROTOCOL STANDARDS

- **CAP v1.2 Protocol** `ITU-T X.1303` — Common Alerting Protocol for emergency public sirens and cell broadcasts
- **IMD / Open-Meteo** `15-min Telemetry` — Real-time rainfall radar feeds and 7-day predictive precipitation curves
- **USGS & NCS Seismology** `Live GeoJSON` — Earthquake epicenter feeds, magnitude thresholds & shake map contours
- **OSRM / OpenStreetMap** `Routing Engine` — Live multi-modal evacuation corridor routing to nearest safe shelters

7. TESTING, QUALITY ASSURANCE & CI/CD

100% Pass Rate

AUTOMATED TESTING & SECURITY VERIFICATION

- **Vitest Test Runner** `v5.0.3` — 192 unit and integration tests covering scoring, auth & geospatial math
- **PyTest Suite** `v8.0.0+` — Backend automated test suite verifying FastAPI routes & PostGIS schemas
- **Secret Guard Scanner** `check-secrets` — Pre-build zero-leakage scanner checking 98 source files for credentials
- **Web Vitals Telemetry** `v6.2.3` — Continuous production tracking of LCP, FID, and CLS performance